

Question 1

Steps

1. Formula:

$$h_{\text{steps}}(s) = m(s, t)$$

s = current word, t = goal word

$m(s, t)$ = number of positions where the two words have different letters

2. Unavoidable remaining cost

Every position that differs from the goal must be changed at least once. Since each action costs 1, a word with m mismatched positions requires at least m additional units of cost.

3. Overestimate

It can't overestimate because no solution can fix m differing positions in fewer than m moves. Dictionary restrictions may force extra moves, but it can't be less than m, so the heuristic is at most the true optimal remaining cost.

Scrabble

1. Formula:

$$h_{\text{scrabble}}(s) = \text{sum of } V(t_i) \text{ for all indices } i \text{ where } s_i \neq t_i$$

2. Unavoidable remaining cost

Every mismatched position must receive its corresponding goal letter. Each insertion costs that goal letter's assigned Scrabble value, and the sum of those values is the cost that must be paid.

3. Overestimate

It's not possible to overestimate because the heuristic counts only the necessary goal-letter insertion. Each occurs separately because only one action can be done at a time to change a position. Any intermediate costs are also not part of the heuristic, so the estimate cannot exceed the cheapest remaining path's cost.

Frequency

1. Formula:

$$h_{\text{frequency}}(s) = 0 \text{ if } s = t, \text{ else } m(s, t) + R(t) \text{ when } s \neq t$$

2. Unavoidable remaining cost

For a non-goal state with m mismatched positions, every solution must pay at least m base move charges of 1. It must also enter the goal word, paying the goal's rarity value, so the cost at minimum is $m + R(\text{goal})$.

3. Overestimate

It cannot be overestimated because any solution uses at least m moves and includes the goal's rarity charge. All other rarity changes are nonnegative, so ignoring them does not make the estimate larger; therefore, $m + R(\text{goal})$ cannot exceed the true optimal remaining cost.

Question 2

Word pair	Cost measure	UCS cost	UCS nodes expanded	A* cost	A* nodes expanded	Reduction
WHY → ASK	Steps	5	773	5	16	97.93%
WHY → ASK	Scrabble	9	765	9	13	98.30%
WHY → ASK	Frequency	38.85540652	808	38.85540652	634	21.53%
BEAR → DUCK	Steps	4	691	4	6	99.13%
BEAR → DUCK	Scrabble	11	2,981	11	6	99.80%
BEAR → DUCK	Frequency	38.2402284	1,084	38.2402284	302	72.14%

1. Yes, UCS and A* returned identical optimal costs in all cases. Example: WHY → ASK: 5 for steps, 9 for Scrabble, and 38.855 for frequency.
2. Scrabble produced the largest percentage reduction for both word pairs. For the why/ask pair, it went from 765 to 13, and for the bear/duck pair, it went from 2981 to 6.
3. Frequency was the least informative for both word pairs. For the why/ask pair, the reduction was 21.53%, and for the bear/duck pair was 72.14%.
4. Steps and Scrabble heuristics give useful estimates of the replacements needed. Scrabble also includes the cost of inserting the required goal letters, helping A* avoid paths that look cheap but are actually pretty expensive to finish. The frequency heuristic ignores the rarity costs of intermediate words, and its estimates leave out much of the remaining cost, giving A* less guidance.

Question 3

Steps - Yes, because every move costs 1 in either direction. Reversing a path preserves the same number of moves, so the shortest path in both directions has the same cost.

Scrabble - No, because the cost depends on the newly inserted letter, which may be different when the move is reversed. An example would be going from dog to jog. J costs 6 and d costs 2.

Frequency- No, because the cost depends on the word entered, and the starting word is not charged. For example, to go from dog to job, it is 11.05 because of $1 + \text{rarity}(\text{jog})$; when reversing, it would be 6.819 because it goes $1 + \text{rarity}(\text{dog})$.

Question 4

In DOOGAL, the initial and goal states are lowercase words of the same length—three or four letters—and every legal action changes exactly one letter. Under the frequency objective, entering word w costs $1 + \text{rarity}(w)$, where the nonnegative rarity value is stored in the dictionary.

The baseline heuristic $d(n)$ is the number of positions in which state n differs from the goal. Propose a rarity-aware heuristic intended to be more informative than d while remaining admissible. Define it precisely and justify your proposal.

By "more informative," we mean that the proposed heuristic should be at least as large as d for every state and strictly larger for at least one state, without overestimating the true optimal remaining cost.

Done with ...

A clean rarity-aware heuristic is to keep the Hamming-distance lower bound, then add rarity costs that every possible solution path must pay.

Let

$$d(n) = h_0(n) = \# \{ i : n_i \neq \text{goal}_i \}$$

and let

$$r_{\min} = \min_{w \in D} \text{rarity}(w)$$

where D is the supplied dictionary.

Define

$$h_R(n) = \begin{cases} 0, & n = \text{goal} \\ d(n) + \text{rarity}(\text{goal}) + (d(n) - 1)r_{\min}, & n \neq \text{goal} \end{cases}$$

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2. $h_R(n) = 0$ if $n = \text{goal}$, else $d + R(\text{goal}) + (d - 1)r_{\min}$ when $n \neq \text{goal}$
3. Yes, the heuristic is admissible. Any solution needs at least d moves. The final move pays the goal's rarity, and the other required moves each pay at least $r_{\min}$. Therefore, the heuristic counts only unavoidable costs and cannot overestimate.
4. It does not need a repair. The heuristic is valid. It must return 0 at the goal because no further costs are needed.
5. With the baseline and proposed heuristics, A* returned the same costs: 38.8554 for the why/ask pair and 38.24022 for the bear/duck pair. The proposed heuristic reduced expansions from 755 to 634 and from 720 to 301. These results support the proposed heuristic.

AI use: For this assignment, I used ChatGPT Astra to understand the code and what some of the questions were asking.